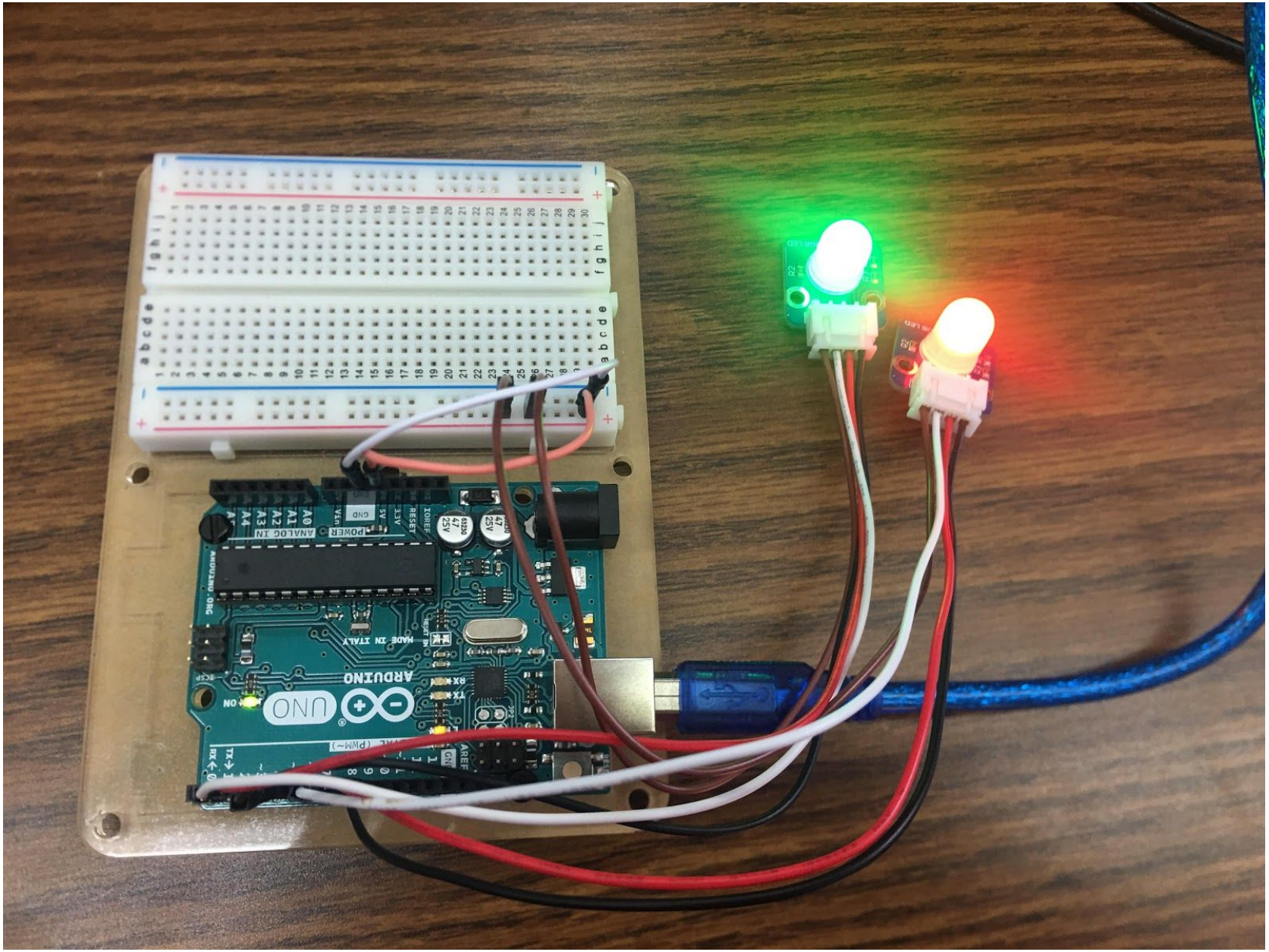
```
/*TITLE: ACTIVITY TRAFFIC LIGHT
 * CREATED BY DONIYOR NIMATULLO
 * STARTED AND FINISHED 11/20/18
 * DESCRIPTION: USE FUNCTIONS TO SIMULATE A TWO WAY INTERSECTION WITH 2 TRAFFIC LIGHTS
 */
int red1 = 2, green1 = 3, blue1 = 4, red2 = 7, green2 = 8, blue2 = 9;
void setup()
{
  Initialize();
}

void loop()
{
  TrafficProtocol();
}

void Initialize()
{
  pinMode(red1, OUTPUT);
  pinMode(green1, OUTPUT);
  pinMode(blue1, OUTPUT);
  pinMode(red2, OUTPUT);
  pinMode(green2, OUTPUT);
  pinMode(blue2, OUTPUT);
}

void TrafficProtocol()
{
  analogWrite(red1, 255); //starts off as green
  analogWrite(green1, 0);
  analogWrite(blue1, 255);

  analogWrite(red2, 0);
  analogWrite(green2, 255); //starts off as red
  analogWrite(blue2, 255);
  delay(2000);

    while (red1 == 255);
    {
      analogWrite(red1, 0); //changes to yellow (from green)
      analogWrite(green1, 0);
      analogWrite(blue1, 255);
      delay(2000);

      analogWrite(red1, 0);
      analogWrite(green1, 255); //changes to red (from yellow)
```

```
{
  analogWrite(red1, 255); //starts off as green
  analogWrite(green1, 0);
  analogWrite(blue1, 255);

  analogWrite(red2, 0);
  analogWrite(green2, 255); //starts off as red
  analogWrite(blue2, 255);
  delay(2000);

    while (red1 == 255);
    {
      analogWrite(red1, 0); //changes to yellow (from green)
      analogWrite(green1, 0);
      analogWrite(blue1, 255);
      delay(2000);

      analogWrite(red1, 0);
      analogWrite(green1, 255); //changes to red (from yellow)
      analogWrite(blue1, 255);

      analogWrite(red2, 255); // sets other light to red
      analogWrite(green2, 0);
      analogWrite(blue2, 255);
      delay(2000);

      while (red2 == 255);
      {
        analogWrite(red2, 0); //changes to yellow (from green)
        analogWrite(green2, 0);
        analogWrite(blue2, 255);
        delay(2000);

        analogWrite(red2, 0);
        analogWrite(green2, 255); //changes to red (from yellow)
        analogWrite(blue2, 255);

        analogWrite(red1, 255); // sets other light to red
        analogWrite(green1, 0);
        analogWrite(blue1, 255);
        delay(2000);
      }
    }
  }
```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